
Breaking through the Tourism Dilemmas in Juneau: The Road to Sustainable Development Driven by a Multi - objective Model Summary

This study focuses on developing a **sustainable tourism model for Juneau, Alaska**, which is facing numerous problems caused by overtourism. In 2023, Juneau received 1.6 million cruise passengers, generating \$375 million in revenue. However, it also encountered issues such as glacier retreat, strained infrastructure, and social discontent. A **multi - objective optimization model** is constructed in this study, taking into account factors like the number of tourists, economic income, hidden costs, environmental protection, and measures to stabilize the tourism industry. The model aims to maximize tourism revenue, minimize ecological and environmental pressure, and enhance the overall satisfaction of residents and tourists. Decision variables include the number of tourists, government restrictions on numbers and consumption, tourism tax, tourist fees, and various government investments in infrastructure, environmental protection, and promotion. The constraints cover aspects such as tourist number limits, glacier protection, resource carrying capacity, and minimum - standard requirements.

The **gradient descent method** is used for data processing and fitting in the research, and the **Non - dominated Sorting Genetic Algorithm II (NSGA - II)** is employed to solve the model, accompanied by visualization processing. A comprehensive optimal solution is selected from the obtained results. The optimal solution shows that receiving **12,000 tourists per day**, imposing **an 8% tourism tax**, charging **a \$15 tourist fee per person**, and allocating **30% of the revenue from tourism tax and tourist fees to environmental protection, 40% to infrastructure construction, and 20% to tourism promotion** can achieve the sustainable development of the tourism industry. Sensitivity analysis indicates that the number of tourists, alcohol consumption restrictions, infrastructure construction, and key coefficients have a significant impact on the model results, and attention should also be paid to investment in environmental protection.

This model can be applied to other tourist destinations. For regions with high ecological pressure like Hawaii, the focus of adjustment lies on the environmental impact part of the model. Through model adjustment and calculation, it can be concluded that more emphasis should be placed on the ecological protection of this region. For areas with weak infrastructure, such as Yosemite National Park, strengthening infrastructure construction is the key. In areas with unique cultures, like New Orleans, cultural factors need to be considered. For regions with unbalanced tourism resource development, such as New York State, publicity and tourism resource development are crucial. Finally, a **memo** is submitted to the Juneau Tourism Committee, elaborating on the research findings, the effects of various measures, and suggestions for optimizing tourism outcomes, aiming to **balance** economic growth, environmental protection, and social well - being in the development of Juneau's tourism industry.

Keywords: Sustainable tourism; Juneau; Multi - objective optimization; NSGA - II